Gen AI for Front Office Analytics

Fundamentals Modelling & Innovation Team

Trading Analytics & Insights | BP Trading

Tools, Use Cases & Innovation Day Preparation

2025–2026

INTRO CONTEXT

Why Now — The AI Productivity Shift

2023: Experimentation

ChatGPT launched. Teams explored what AI could do. Most usage was ad hoc and unstructured — copy-paste into chat windows.

2024: Tool Integration

AI moved into IDEs and workflows. GitHub Copilot, Cursor, Claude Code became daily drivers. Measurable productivity gains emerged.

2025: Agentic & Enterprise

AI agents run autonomously. Multi-agent orchestration, enterprise governance, and Copilot Studio transform business processes.

2026: What We're Building

Front office teams that master these tools now will have compound advantages. Innovation Day is our launchpad.

The window to build AI fluency is now — teams that wait will fall behind teams that experiment

INTRO GOALS

What We'll Cover Today

1. What tools are available now (Microsoft Copilot, GitHub Copilot, Cursor, Claude Code, Copilot Studio) and use cases for each
2. How to access these tools — licensing, request process, and getting set up in your workflow
3. Ethics and compliance considerations — latest guidance and where to find it
4. Prepare for Innovation Day — explore commercial use cases to enhance front office productivity
5. Populate an ideas board of current uses and new commercial ideas to pursue

Focus: entirely on how to USE these tools to unlock value — not how LLMs work under the hood

INTRO | FM&I

Who We Are — FM&I Team Context

Fundamentals Modelling & Innovation

- Build, maintain and continuously improve fundamental models supporting T&S commercial decisions
- Cross-discipline squads collaborating with traders, market strategists, and analytics partners
- Full data ELT pipelines on Dataiku — automated projects with scenarios and triggers
- Ad hoc data analysis for trading desks plus statistical/ML model development for predictions
- Advancing capabilities using AI/ML, Dataiku automation, and advanced analytics

How Gen AI Maps to Our Daily Work

ELT Pipeline Development	Cursor / Copilot	2-3x faster pipeline builds
Ad Hoc Data Analysis	Claude Code	Minutes instead of hours
Model Development	Copilot + Cursor	Faster prototyping & validation
Code Review & Debug	Cursor Composer	Multi-file trace & fix
Documentation	Copilot Ask	Auto-generate model docs
Dashboard Prototyping	Claude Code	Plotly apps in minutes

Every task in our daily workflow has an AI tool that can accelerate it — the question is which one and how

02

Current Tool Landscape

What's available, what each tool does, and when to use which one

TOOLS CODING

AI Coding Tools — Detailed Comparison

GitHub Copilot

IDE Plugin (VS Code, JetBrains, Neovim)

$19–39/mo

Features: Inline autocomplete (Tab to accept), Copilot Chat sidebar, Agent Mode (multi-file), Edit Mode (inline diffs), Workspace Indexing, CLI suggestions, multi-model (GPT, Claude, Gemini)

Best for: Day-to-day coding speed, inline suggestions while typing, teams already in VS Code. Lowest friction — stays in your flow.

Key differentiator: Most widely adopted. Now supports multiple AI models. Agent mode catching up to Cursor. Strong GitHub ecosystem integration.

Cursor

AI-Native IDE (VS Code fork)

$20/mo Pro

Features: Composer (multi-file agent), Tab autocomplete, Chat, .cursorrules project files, Background Agents, Agent Teams, MCP integration, Git Worktrees, Bug Finder

Best for: Complex multi-file tasks, project-level refactoring, teams wanting AI-first IDE. Composer agent is the standout feature.

Key differentiator: Purpose-built for AI workflows. .cursorrules gives team-wide AI consistency. Background agents for async work. Most advanced agent capabilities.

Claude Code

CLI Agent (Terminal-based)

Pro/Max plan

Features: Full shell access, parallel agents, sub-agents, agent teams, skills system, hooks, memory across sessions, MCP servers, Claude Code for Web (browser-based)

Best for: Complex builds from scratch, exploring unfamiliar codebases, automation pipelines, tasks requiring shell/filesystem access. Power users.

Key differentiator: Only CLI tool — runs in terminal, not IDE. Deepest system access. Agent teams for parallel work. Skills + hooks for automation. Most agentic.

GitHub Copilot = widest adoption | Cursor = deepest agent features | Claude Code = most powerful CLI tool

TOOLS ENTERPRISE

Enterprise & Office AI Tools

Microsoft 365 Copilot

AI embedded across the Office suite

$30/user/mo

Features: Drafts in Word, analysis in Excel, summaries in Teams, email triage in Outlook, presentations in PowerPoint, Copilot Chat across all M365 data via Microsoft Graph

Best for: Knowledge workers who live in Office apps. Meeting summaries, email management, document drafting, data analysis. Biggest time savings for non-coding tasks.

Key differentiator: Only tool that works INSIDE your existing Office apps with your organizational data. Microsoft Graph gives it access to your emails, files, meetings, and chats.

Copilot Studio

No-code to pro-code agent platform

Consumption

Features: Visual agent builder, multi-agent orchestration, 1,400+ connectors, MCP server support, Computer Use (preview), GPT-5/4.1 + third-party models, enterprise governance (Entra, Defender, DLP)

Best for: Building automated business workflows, customer-facing bots, internal helpdesks, autonomous agents triggered by events. IT and business teams building without code.

Key differentiator: The enterprise agent PLATFORM — not a single tool but a factory for building, deploying, and governing agents at scale. Connects to almost any enterprise system.

Copilot for Power BI

AI-powered data analysis in dashboards

Included w/ M365

Features: Natural language queries on dashboards, auto-generate DAX formulas, create visuals from prompts, explain trends, identify outliers, narrative summaries of data

Best for: Analysts and traders who consume dashboards. Ask questions in plain English instead of navigating complex filters. Democratizes data access across the organization.

Key differentiator: Turns Power BI from a 'look at data' tool into a 'talk to data' tool. Traders can query commodity analytics conversationally without learning DAX or building reports.

M365 Copilot = AI in your Office apps | Studio = build your own agents | Power BI = talk to your dashboards

TOOLS FRAMEWORK

The Simple Rule — Which Tool, When

Don't overthink it. Start with this decision tree and you'll pick the right tool 90% of the time.

Are you writing code line-by-line in an IDE?	**GitHub Copilot** Tab autocomplete, inline suggestions, lowest friction
Do you need to change multiple files at once?	**Cursor Composer** Multi-file agent, understands project structure, background agents
Are you building something from scratch or exploring a repo?	**Claude Code** Full shell access, parallel agents, natural language → complete solution
Are you working in Office apps (email, docs, meetings)?	**M365 Copilot** Embedded in Teams, Outlook, Word, Excel, PowerPoint natively
Do you need to automate a business process?	**Copilot Studio** No-code agent builder, 1,400+ connectors, enterprise governance

Match the tool to the task — not the other way around. When in doubt, start with Ask mode in any tool.

TOOLS GITHUB COPILOT

GitHub Copilot — Deep Dive

Key Features

- Inline Completions — real-time code suggestions as you type, Tab to accept
- Copilot Chat — conversational AI in sidebar, ask questions about your codebase
- Agent Mode — multi-file autonomous coding, creates and edits files to complete tasks
- Workspace Indexing — understands your full repo structure for context-aware suggestions
- Ask Mode — explain code, generate docs, answer questions without making changes
- Edit Mode — targeted refactoring with inline diff preview before accepting
- Copilot for CLI — terminal command suggestions and explanations

Modes Explained

Ask

Read-only Q&A about your code. No file changes. Best for understanding and learning.

Edit

Targeted changes with inline diffs. You review each change before accepting.

Agent

Autonomous multi-file editing. Plans steps, creates files, runs commands. Most powerful.

55% faster coding reported by GitHub Copilot users • 30-35% suggestion acceptance rate

TOOLS IN ACTION

See It in Action — The Autocomplete Effect

You're writing a function to calculate rolling correlations between Brent and WTI:

What you type (3 lines):

```
def rolling_correlation(brent_df, wti_df,
    window=20):
    """Calculate rolling correlation..."""
```

What Copilot autocompletes (15 lines):

```
    merged = brent_df.join(wti_df, how='inner')
    merged = merged.dropna()
    corr = merged.iloc[:,0].rolling(window)\
        .corr(merged.iloc[:,1])
    return corr.dropna()  # + null handling, etc.
```

The math that matters

3 lines → 15 lines

Copilot wrote 80% of the function from your context

3 minutes saved

on ONE function — seems small

× 50 functions/day

you write dozens of similar utility functions daily

= 2.5 hours saved

per day, just from autocomplete — no agent mode needed

It's not about one function — it's the compound effect across hundreds of small moments per day

TOOLS CURSOR

Cursor — Deep Dive

Key Features

- Composer — multi-file agent that plans and edits across your entire project
- Tab Autocomplete — context-aware inline suggestions (like Copilot but in Cursor IDE)
- Chat — conversational AI with full codebase awareness via indexing
- .cursorrules — project-level instructions that shape AI behavior for your team
- Background Agents — run long tasks in background while you keep working
- Agent Teams — multiple AI agents collaborating on different parts of a task
- MCP Integration — connect to external tools and data sources via Model Context Protocol
- Git Worktrees — work on multiple branches simultaneously with separate agent contexts

What Makes Cursor Different

AI-Native IDE

Built from scratch around AI — not a plugin bolted onto an existing editor. Every feature designed for AI-first workflows.

Composer Agent

The killer feature. Describe what you want in natural language, Composer plans the approach and edits files across your project.

Project Rules

.cursorrules files let you define coding standards, patterns, and context that persist across sessions for your whole team.

Background Agents

Kick off a task, keep coding. Background agents work in parallel on separate branches and report back when done.

Cursor is the power tool for multi-file, project-level AI coding — Composer is the game changer

TOOLS IN ACTION

See It in Action — Multi-File Refactor

You type this into Cursor Composer:

```
Refactor signal_generation.py: separate data fetching, signal calculation, and risk checks into 3 files. Add type hints and docstrings. Update all imports.
```

What Cursor does

1. Creates 3 new files: data_fetch.py, signal_calc.py, risk_checks.py
2. Moves the right code into each file with proper structure
3. Updates 7 import statements across the project
4. Adds type hints to 23 functions
5. Writes docstrings for every function and class
6. Shows you the complete diff to review before accepting

45 seconds

11 files edited

from one sentence

Manual estimate:

2–3 hours of careful work

This is where AI coding goes from 'nice autocomplete' to 'fundamentally different way of working'

TOOLS HEAD TO HEAD

GitHub Copilot vs Cursor

GitHub Copilot

Pros

- Most widely adopted — largest community, most integrations
- Works in VS Code, JetBrains, Neovim, Xcode — not tied to one IDE
- Multi-model support: GPT, Claude, Gemini — choose per task
- Deep GitHub integration: Issues → PRs, Coding Agent, Actions
- Enterprise plan with SSO, audit logs, IP indemnity, policy controls
- Free tier available — low barrier to start experimenting

Cons

- Agent mode less mature than Cursor Composer — catching up
- No background agents or async task execution yet
- No project-level rules file equivalent to .cursorrules
- Inline suggestions can be distracting for some workflows

Cursor

Pros

- Composer agent is best-in-class for multi-file autonomous coding
- Background agents — async tasks on separate branches while you work
- .cursorrules files — team-wide AI configuration shared via git
- Agent Teams — multiple AI agents collaborating on a single task
- MCP integration — connect to external tools and data sources natively
- AI-native design — every feature built around AI-first workflows

Cons

- Locked to Cursor IDE — can't use in JetBrains, Neovim, etc.
- Smaller community and ecosystem than GitHub Copilot
- No free tier — $20/mo minimum for meaningful features
- VS Code fork means occasional lag behind VS Code updates

Recommendation: Use Copilot for daily coding speed + Cursor for complex multi-file tasks — they complement each other

TOOLS

CURSOR ADVANCED

Cursor — Advanced Features

Background Agents

Kick off a task in the background — Cursor works on it in a separate branch while you keep coding. Check progress anytime. When done, review the diff and merge. Like having a junior developer working in parallel.

How to use: Click 'Background Agent' in Composer, describe the task, Cursor creates a branch and works autonomously. You get a notification when it's done.

Agent Teams (Preview)

Multiple AI agents collaborate on different parts of a task. One handles the frontend, another the backend, a third writes tests. They coordinate through shared context — not just parallel execution.

How to use: Currently in preview for Cursor Business. Define team roles in your prompt: 'Spawn agents for API, UI, and tests — coordinate through shared interfaces.'

Git Worktrees

Work on multiple branches simultaneously, each with its own Cursor agent context. Perfect for: fixing a bug on one branch while building a feature on another, without losing AI context.

How to use: Use Git worktrees (git worktree add) to create separate working directories. Open each in Cursor — each gets independent AI context and can run background agents.

.cursorrules Deep Dive

Project-level AI instructions that persist across sessions and team members. Define coding standards, architecture patterns, domain context, and tool preferences. Shared via git — the whole squad benefits.

How to use: Create .cursorrules in project root. Include: language prefs, coding standards, domain terms (commodity names, model types), data patterns, test requirements.

These features turn Cursor from a smart editor into a development platform — learn them and your throughput multiplies

TOOLS | CLAUDE CODE

Claude Code & Cowork — Deep Dive

Claude Code (CLI)

Terminal-based agentic coding tool. Unlike IDE plugins, it runs directly in your terminal with full shell access. Read, write, and execute code across your entire project. Parallel agents work on multiple tasks simultaneously. Full repo understanding without indexing delays.

Claude Cowork (Desktop)

Takes Claude Code beyond IDEs. Desktop tool for non-developers to automate file and task management. Manages files, processes documents, creates reports — all via natural language. Bridges the gap between coding tools and day-to-day productivity for the whole team.

CLI vs IDE — Why It Matters

Aspect	CLI (Claude Code)	IDE (Copilot/Cursor)
Context	Full filesystem + shell access	Project workspace only
Execution	Runs scripts, installs packages, tests	Suggests code, you run it
Workflow	Natural language → complete solution	Code alongside, accept/reject
Multi-task	Parallel agents on separate tasks	One conversation at a time
Access	Terminal — works anywhere SSH works	Requires GUI application
Best for	Complex builds, exploration, automation	Daily coding, refactoring

Claude Code = power tool for complex tasks • Cowork = AI productivity for everyone, not just coders

TOOLS IN ACTION

See It in Action — Prompt to Dashboard

The prompt you type into Claude Code:

```
Build a Plotly Dash app: Brent forward curve from our CSV, historical vs implied vol surface, cross-commodity correlation heatmap. Add date selector + PDF export.
```

What Claude Code does next

1. Reads your CSV files to understand the data structure
2. Writes ~400 lines of Python (Dash layout + Plotly figures)
3. Creates 3 interactive chart components with callbacks
4. Adds the date range selector and PDF export button
5. Installs dependencies, runs the app, opens browser
6. Iterates if anything breaks — fixes its own errors

12 minutes

from prompt to working interactive dashboard

Manual approach:
2–3 days of development

The gap between 'I have an idea' and 'I have a working tool' just collapsed from days to minutes

TOOLS CLAUDE TEAMS

Claude Code — Agent Teams & Hooks

Agent Teams (Experimental, v2.1.32+)

- Multiple Claude Code instances working in parallel on the same codebase
- Team Lead coordinates: breaks down tasks, assigns work, synthesizes results
- Teammates are full independent sessions — own context window, own tools
- Unlike sub-agents: teammates can message each other directly, not just report up
- Shared task list with dependencies — tasks unblock automatically when prereqs complete
- Delegate Mode: restrict lead to coordination only — forces proper delegation
- Use cases: multi-layer refactoring, parallel code review, competing debug hypotheses
- Requires Opus 4.6 — high token usage, best for large-scale tasks
- Enable: set CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1 in settings

Timeline & Roadmap

- May 2025: Claude Code launched with Opus 4
- Oct 2025: Opus 4.6 + Sonnet 4.6 release
- Jan 2026: v2.1.0 — skills, memory, hooks
- Feb 2026: Agent Teams research preview
- Next: role-based model selection per agent

Claude Code for Web

- Browser-based at claude.ai/code
- Parallel agents in the cloud
- GitHub integration built in
- Teleport-to-local for finishing work
- Available on Pro/Max/Team/Enterprise

Agent Teams = multi-threaded AI development • 'Like going from single-threaded to multi-threaded programming'

TOOLS MODES

Modes — Ask & Edit (You Stay in Control)

Understanding modes is critical. Using the wrong mode wastes time. Ask mode for understanding, Agent mode for autonomous work. Match the mode to the task.

Ask / Chat Mode

Read-only. Ask questions, get explanations.

No files are changed. The AI just answers.

Use when:

Understanding unfamiliar code, learning a library, exploring options before committing to an approach

Copilot Chat | Cursor Chat | claude (ask)

Edit / Inline Mode

Targeted changes with review.

You see diffs before accepting. You approve every change.

Use when:

Small refactors, renaming variables, fixing a specific function, adding error handling to existing code

Edit Mode | Cmd+K inline | —

Ask and Edit modes keep you in control — use them for 60% of your interactions

TOOLS AGENT MODE

Agent Mode — The AI Does the Work

Agent Mode

The most powerful mode across all tools

The AI plans steps, creates and edits multiple files, runs terminal commands, tests its work, and iterates until done. You describe what you want — it figures out how to build it.

Use when: new features, complex bugs across files, project scaffolding, large refactors, building from scratch

GitHub Copilot

Agent Mode

Multi-file editing, terminal execution, error self-healing. Plans approach and executes autonomously.

Cursor

Composer Agent

The killer feature. Plans across your entire project, edits files, runs tests. Background agents work in parallel.

Claude Code

Default mode

Born agentic. Full filesystem + shell access. Parallel agents. The deepest access of any coding AI tool.

Agent mode is where the magic happens — but start with Ask mode until you're comfortable, then graduate up

TOOLS BEYOND CODING

Beyond Coding — Non-Code Workflows

These tools aren't just for writing code. GitHub Copilot, Cursor, and Claude Code can accelerate non-software-engineering tasks that FM&I does every day — from data analysis to documentation to research.

Research & Summarize

Claude Code / Cursor Chat

'Read this 50-page research paper on commodity volatility modelling and summarize the key methodology, findings, and limitations in 500 words'

Data Exploration

Claude Code

'Load this CSV of daily Brent prices, calculate basic statistics, identify structural breaks, and plot the time series with annotated events'

Document Generation

Copilot Ask / Claude

'Generate a model risk document from this codebase following our standard template — inputs, methodology, assumptions, limitations, validation results'

Email & Communication

M365 Copilot / Claude

'Draft a summary email to the trading desk explaining why the nat gas model's predictions diverged from spot prices this week — use attached analysis'

File & Data Management

Claude Code / Cowork

'Organize these 200 CSV files by commodity and date range, merge the ones that overlap, and flag any with missing data or format inconsistencies'

Presentation Creation

Claude Code / Cowork

'Create a 10-slide presentation summarizing this quarter's model performance metrics for the trading desk review'

If it involves text, data, or files — AI coding tools can probably help, even if no code is involved

TOOLS ROADMAP

GitHub Copilot & Cursor — Roadmap

GitHub Copilot — Where It's Heading

- Copilot Coding Agent: assign Issues, get draft PRs — async cloud-based agent
- Multi-model: Claude, GPT, Gemini all available — pick per task
- Agent mode maturing: multi-file edits, terminal execution, self-healing errors
- MCP Registry: growing ecosystem of MCP servers installable with one click
- Custom Agents: define specialized agents via AGENTS.md file
- Copilot Extensions: third-party integrations expanding rapidly
- Vision: from autocomplete tool → full development platform with agent orchestration

Cursor — Where It's Heading

- Background Agents maturing: run complex tasks asynchronously on separate branches
- Agent Teams: multiple specialized agents collaborating on different parts of a task
- Deeper MCP integration: richer tool ecosystem, resource support, one-click setup
- Improved context management: smarter indexing, better long-session handling
- Bug Finder: proactive code analysis that identifies issues before you ask
- Enhanced .cursorrules: more powerful project-level AI configuration
- Vision: from AI IDE → AI-native development platform where agents do the work

Both tools are converging on the same vision: AI agents that own tasks end-to-end, not just suggest code

03

Microsoft Copilot Ecosystem

M365 Copilot, Copilot Studio, Agents, and the Enterprise AI platform

MICROSOFT M365 COPILOT

Microsoft 365 Copilot — Overview

Teams

Meeting summaries, action items, catch-up on missed meetings 4x faster. Real-time transcription and intelligent recap.

Outlook

Draft emails, prioritize inbox, summarize threads, schedule smart. Turns email management from hours to minutes.

Word

Draft documents from prompts or data. Rewrite, summarize, change tone. First-draft quality in seconds.

Excel

Analyze data with natural language. Generate formulas, create charts, identify trends. Now with Anthropic reasoning models.

PowerPoint

Create presentations from prompts or documents. Add slides, restructure, design — AI as creative partner.

Copilot Chat

Cross-app assistant grounded in your M365 data via Microsoft Graph. Ask questions across all your documents and emails.

Internal BP training available — see resources slide for links. Short demo to follow.

70% of users report higher daily productivity • 29% faster task completion • $30/user/month

MICROSOFT IN ACTION

See It in Action — The Meeting You Missed

The scenario

You missed the 2-hour weekly trading strategy review. 47 discussion points, 12 action items, 3 decisions that directly affect your models.

What Copilot gives you

- Full structured summary with key decisions highlighted
- Your 3 action items extracted and listed with deadlines
- The decision to switch nat gas model methodology — flagged as high-relevance to you
- Links to the 2 documents referenced during discussion
- Follow-up questions you can ask Copilot about specific discussion points

90 seconds

to fully catch up

vs. watching a 2-hour recording
or asking 5 colleagues

This isn't about coding — it's about reclaiming hours every week from information overhead

MICROSOFT POWER BI

Copilot for Power BI

Copilot for Power BI transforms Power BI from a 'look at data' tool into a 'talk to data' tool. Ask questions in plain English, get instant visualizations and insights — no DAX knowledge required. For FM&I, this means traders and analysts can query commodity dashboards conversationally.

What It Can Do

- Ask natural language questions: 'What was the Brent-WTI spread last quarter?'
- Auto-generate DAX formulas from plain English descriptions
- Create new visuals and charts from conversational prompts
- Summarize and explain trends: 'Why did revenue drop in March?'
- Identify outliers and anomalies in datasets automatically
- Generate narrative summaries of reports for stakeholder review
- Suggest follow-up questions based on the data you're viewing

FM&I Use Cases

Traders query dashboards	*'Show me Henry Hub vs TTF price ratio for the last 6 months with 20-day rolling average'*
Anomaly investigation	*'Which commodity feeds had the most missing data points this week?'*
Model performance	*'Compare our nat gas forecast accuracy across all models for Q4'*
Ad hoc analysis	*'What's the correlation between Brent crude and USD/EUR for 2025?'*
Stakeholder reports	*'Generate a narrative summary of this quarter's spread performance'*

Copilot for Power BI = natural language access to your dashboards — no DAX, no filters, just ask

MICROSOFT COPILOT STUDIO

Copilot Studio — The Agent Platform

Microsoft Copilot Studio is Microsoft's fully managed, no-code to pro-code platform for building, deploying, and governing AI agents across the enterprise. It integrates with M365, Dynamics 365, Power Platform, and 1,400+ connectors to create agents that answer questions, automate workflows, and take autonomous action.

No-Code Agent Builder

Describe your agent in natural language. Add knowledge, topics, actions — publish in minutes.

Multi-Agent Orchestration

Agents delegate to other agents — Copilot Studio, M365, Azure AI, Fabric agents work together.

Computer Use (Preview)

Agents operate apps and websites with virtual mouse and keyboard — automate where no API exists.

1,400+ Connectors + MCP

SAP, Salesforce, ServiceNow, Databricks, databases, and any MCP server with a few clicks.

Model Choice

GPT-5, GPT-4.1, and third-party models. Pick the right model per agent for performance and cost.

Enterprise Governance

Entra Agent ID, Defender protection, audit logging, DLP, data residency, admin analytics.

IDC projects 1.3 billion AI agents by 2028 • Microsoft 365 Copilot has crossed 100M monthly active users

MICROSOFT AGENTS

Copilot Agents — Types & Maturity

Declarative Agents

Extend M365 Copilot

Lightweight agents in Agent Builder or Copilot Studio. Add custom knowledge and instructions to the base Copilot — like a specialized expert for a domain.

Use cases: HR policy bot, contract Q&A, onboarding assistant

Build: Minutes to hours

Conversational Agents

Custom chat experiences

Full agents with rich dialog flows, generative answers from knowledge sources, and actions connecting to backend systems via connectors and Power Automate.

Use cases: Customer support, order tracking, supplier inquiry, internal KB

Build: Hours to days

Autonomous Agents

Event-driven, self-acting

Run independently — triggered by events (email, threshold, schedule). Execute multi-step workflows, make decisions, escalate when needed. Deep reasoning for complex analysis.

Use cases: Lead qualification, invoice processing, compliance monitoring

Build: Days to weeks

The shift: 'AI answers questions' → 'AI completes workflows' → 'AI runs processes autonomously'

MICROSOFT CONNECTORS

Databricks & Dataiku Connectors

Copilot Studio connects to 1,400+ systems via Power Platform connectors, MCP servers, and custom connectors. For FM&I, the critical integrations are with our data platforms.

Databricks — Native Connector

- Native Power Platform Connector (Public Preview, June 2025) — single-click setup
- Add Databricks as direct knowledge source for agents — select catalog and tables
- MCP Server integration — Unity Catalog functions exposed to agents via OAuth
- Governance preserved — Unity Catalog access controls flow through to Power Platform
- Use cases: query pricing data, trigger ETL, ask natural-language questions on lakehouse

Dataiku — API & Custom Connector

- No native Power Platform connector yet — connect via REST API custom connector
- Build custom connector pointing at Dataiku DSS API → use as tool in Copilot Studio
- Dataiku's own agent framework: Agent Connect, LLM Mesh, LangChain agents
- Native Power BI export plugin — push datasets to Power BI, Copilot accesses via Graph
- Use cases: trigger model scoring, query project outputs, orchestrate ML pipelines

Databricks = plug-and-play today | Dataiku = connect via REST API | Both paths work for FM&I

MICROSOFT USE CASES

Enterprise Use Cases — Copilot Studio

Sales

Sales Development Agent qualifies leads, nurtures pipeline, personalizes outreach — connects to Dynamics 365 and Salesforce.

Qualified opportunities +2.7%, win rates +2.5%

Customer Service

Conversational agents handle tier-1 support with generative answers. Auto-escalate complex issues with full context.

Deflects 40-60% of routine tickets

HR & Onboarding

Employees ask policy questions, submit PTO, check benefits. New hire onboarding automated end-to-end.

Reduces onboarding time by 25%

IT Helpdesk

Password resets, VPN troubleshooting, software requests — agent executes Power Automate flows to resolve.

Resolves 35-50% of tickets autonomously

Finance

Autonomous agents monitor invoice thresholds, flag anomalies, route approvals. Deep reasoning on spend patterns.

Commercial Bank of Dubai: 39K hours saved/yr

Compliance

Monitor regulatory changes, scan contracts for risk clauses, alert teams. Computer Use automates legacy portals.

Continuous monitoring replaces periodic reviews

Pre-built agents ship ready to deploy — or build your own in minutes with Copilot Studio

MICROSOFT DEMO

Creating a Copilot Agent — Example

1. **Describe Your Agent**
   Start in Agent Builder or Copilot Studio. Describe what your agent should do in natural language: 'An agent that helps traders check commodity price feeds and flag anomalies.'

2. **Add Knowledge Sources**
   Connect your data: SharePoint docs, Databricks tables, internal wikis, uploaded files. The agent uses these to ground its responses in your actual data.

3. **Add Actions & Connectors**
   Give it the ability to act: Power Automate flows, connectors to Databricks, email notifications, Teams messages. These are the agent's 'hands'.

4. **Refine Instructions**
   Key technique: ask Copilot to improve its own instructions. 'Review these instructions and suggest improvements for accuracy and edge cases.' Iterate until crisp.

5. **Test, Publish, Govern**
   Test in the built-in test pane. Publish to Teams, SharePoint, or WhatsApp. IT manages via Entra Agent ID and admin analytics.

Pro tip: use the LLM to generate and refine agent instructions — it's better at writing its own prompts

MICROSOFT IN ACTION

See It in Action — Agent Answers a Trader

The scenario: a trader messages the agent in Teams

```
"What was the average Brent-WTI spread last week and how does it compare to the 30-day average?"
```

What the agent does (behind the scenes)

1. Receives the message via Teams connector
2. Queries Databricks via native Power Platform connector
3. Runs SQL: calculates 7-day avg and 30-day avg spreads
4. Formats natural language response with the numbers
5. Sends reply back in the Teams chat — with a chart

The result

8 seconds

from question to answer

Previously: open dashboard, navigate filters, calculate manually, or message the analytics team and wait hours.

Agents don't just answer questions — they eliminate the friction between 'I wonder...' and 'now I know'

MICROSOFT TIMELINE

Copilot — The Pace of Change

Mar 2025	**Autonomous agents go live**	+ deep reasoning, MCP support, generative orchestration
Jun 2025	**Databricks connector launches**	+ Copilot Studio in VS Code
Jul 2025	**WhatsApp + SharePoint channels**	+ MCP one-click connector setup
Sep 2025	**Computer Use preview ships**	+ file upload, analytics themes, MCP resources
Oct 2025	**GPT-4.1 becomes default**	+ automated agent evaluation, express mode
Nov 2025	**GPT-5 Chat goes GA**	+ Entra Agent ID, Agent 365, Defender
Dec 2025	**GPT-5.2 + Anthropic in Excel**	+ Sales Agent, Agent Mode in Office

A major update every month — the platform you see today will be significantly different in 6 months

MICROSOFT ROADMAP

Copilot — What's Coming in 2026

Copilot Tuning

Fine-tune and train models using your specific organizational data. Agents always have access to relevant, domain-specific information without prompt engineering.

Work IQ for Custom Agents

Tap into Microsoft's intelligence layer to build agents tuned for your unique workflows. Respects existing permissions, sensitivity labels, compliance controls.

Agent Factory

Build agents using Microsoft Foundry and Copilot Studio, deploy anywhere including M365 Copilot, with a single metered plan — no upfront licensing.

Expanded Computer Use

Broader availability of agents that operate apps and websites with virtual mouse and keyboard. Moving from US-only preview to global availability.

Agent Mode in Office Apps

'Vibe working' in Word, Excel, PowerPoint — create world-class artifacts by describing what you want. Excel now supports Anthropic reasoning models.

Multi-Agent at Scale

Microsoft Agent 365 as unified control plane — centralized governance, policy management, and monitoring across all agent types and platforms.

2026 theme: from 'AI assists individuals' to 'AI runs business processes' — governed, scalable, enterprise-grade

04

FM&I Use Cases & Workflows

How these tools map to your daily work in commodity trading data science

FM&I

MORNING

Morning — Market Check & Data Prep

8:30 AM Overnight price anomaly check

Claude Code

'Flag anomalies in overnight Brent, WTI, Henry Hub, and metals feeds' — scans your pipelines and summarizes anything unusual before you've finished your coffee.

9:00 AM Update pricing model features

Copilot

Autocompletes pandas/NumPy transformations as you type. Suggests feature engineering patterns based on your existing code. Fastest for incremental changes.

10:00 AM Debug broken ETL pipeline

Cursor

Paste the error into Composer — it traces the issue across ingestion, transform, and load files automatically. Multi-file debugging without switching tabs.

11:30 AM Explore new volatility model

Claude Code

'Explain this repo and summarize the key modeling decisions' — perfect for unfamiliar academic code or new research implementations.

Morning = market check, data prep, and debugging — the AI handles the grunt work so you focus on insights

FM&I

AFTERNOON

Afternoon — Build & Deliver

1:00 PM **Write backtesting harness** — Copilot Agent

Describe test scenarios in natural language. Agent scaffolds the full pytest suite with fixtures for historical commodity data. You review and refine.

2:30 PM **Refactor signal generation code** — Cursor

Multi-file refactor across signal, risk, and execution modules via Composer. It understands the dependencies between files and refactors consistently.

3:30 PM **Document model for compliance** — Copilot Ask

'Generate model documentation covering inputs, assumptions, and limitations' — turns your codebase into a structured risk document in minutes.

4:30 PM **Prototype spread dashboard** — Claude Code

'Build a Plotly Dash app showing crack spread, time spread, and location basis' — from prompt to working interactive dashboard in under 30 minutes.

Afternoon = build, refactor, document, deliver — 2-3 more analyses per week without working longer hours

FM&I

PROMPTS

Prompt Recipes — Commodities

Data Wrangling	*'Write a pandas pipeline that joins ICE Brent futures with EIA inventory data, forward-fills missing dates, and calculates rolling 20-day correlation'*
Model Development	*'Implement a mean-reversion signal for nat gas calendar spreads using a half-life estimator. Include walk-forward validation with 252-day training window'*
Risk & Compliance	*'Generate a model risk document for this PnL attribution model covering: data lineage, key assumptions, sensitivity analysis, known limitations'*
Visualization	*'Create an interactive Plotly dashboard: term structure curves, historical vs implied vol surface, cross-commodity correlation heatmap'*
Code Quality	*'Review this backtest for look-ahead bias, survivorship bias, and data snooping. Flag issues and suggest fixes'*
Legacy Code	*'Explain this 2,000-line R script that calculates VaR for our crude oil book. Identify methodology, key parameters, hardcoded assumptions'*

Key principle: The more domain context — contracts, frequencies, methodologies — the better the output

FM&I

IMPACT

Hours to Minutes — Build & Migrate

Build commodity curve construction script

1–2 days → **2–3 hours**

70%

Backtest a new signal across 10 commodities

3–5 days → **1 day**

60–75%

Migrate pricing model from R/MATLAB to Python

1–2 weeks → **2–3 days**

70%

The biggest savings come from tasks that involve boilerplate, repetition, or cross-referencing large codebases

FM&I

IMPACT

Hours to Minutes — Debug & Deliver

Generate model validation documentation

1–2 days → **2–3 hours**

80%

Debug data quality issue across 5 vendor feeds

2–4 hours → **20 min**

85%

Prototype position monitoring dashboard

3–5 days → **1 day**

70%

Pick one scenario from these two slides and try it this week — time yourself before and after

FM&I

REAL IMPACT

Before & After — A Real FM&I Week

Monday	Debug ETL failure across 3 pipeline stages	4 hours → **30 min**	Cursor
Tuesday	Build new feature for pricing model	Full day → **3 hours**	Copilot Agent
Wednesday	Ad hoc analysis request from trading desk	2 hours → **20 min**	Claude Code
Thursday	Model documentation for compliance review	Full day → **2 hours**	Copilot Ask
Friday	Prototype dashboard for new spread analysis	Full day → **2 hours**	Claude Code

Without AI: barely finished, worked late twice, no time for innovation. **With AI:** done by Wednesday — Thursday and Friday free for the signal research you've been wanting to do.

The Compound Effect

It's not about one task being faster.

It's about EVERY task being faster, ALL DAY, EVERY DAY.

15–30 min	saved per AI interaction
10–15x	interactions per day
2–4 hours	saved daily
40–80 hours	saved per month = an extra work-week

You're not replacing your job. You're giving yourself a second analyst who works at machine speed.

SKILLS PROMPTING

How to Prompt Effectively

Be Specific with Domain Context

Mention specific contracts (ICE Brent, Henry Hub), frequencies (daily, 5-min), methodologies (half-life, GARCH). Generic prompts get generic answers.

Use the LLM to Write Its Own Prompt

Ask: 'I want to build X. Write me the best prompt to give you to achieve this.' Then use that improved prompt. LLMs are better at prompting themselves.

Provide Examples

Show input/output examples. 'Here is a sample of the data. Here is what the output should look like.' One example is worth a thousand words of description.

Request Step-by-Step Reasoning

For complex analytical tasks, ask the AI to think through the problem before coding. This produces better, more correct solutions.

Specify Output Format

State exactly what you want: 'Return a pandas DataFrame with columns X, Y, Z' or 'Create a function that takes X and returns Y'. Reduce ambiguity.

Iterate, Don't Restart

Build on previous outputs. 'Good, now add error handling' or 'Refactor this to use vectorized operations'. The conversation is the context.

The quality of your prompt determines the quality of the output — invest 30 seconds more upfront to save 30 minutes

SKILLS TEMPLATES

Template Projects — .cursorrules & Beyond

Cursor: .cursorrules File

A .cursorrules file lives in your project root and tells Cursor how to behave. It persists across sessions and is shared with your team via git.

Example content:

```
- You are a commodity trading data scientist
- Use pandas, numpy, scipy for analysis
- All datetime handling uses UTC
- Use Plotly for all visualizations
- Follow BP coding standards
- Write docstrings for all functions
- Include type hints
- Test data must use realistic commodity prices
```

Claude: claude.md / CLAUDE.md

Claude Code and Claude Projects use a CLAUDE.md file (or project instructions) to set persistent context.

Example: _p-presentation-creator project

This project (used to build this deck!) has a claude.md with:
- Project structure and file locations
- Design system (colors, fonts, layouts)
- Code generation patterns
- QA and verification workflows
- Reusable component library

Key insight: Investing time in project setup files pays off exponentially across every future interaction.

Template projects with rules files are the highest-leverage investment you can make in AI productivity

MODELS PRIMARY

The Models You'll Use Most

Claude Sonnet 4.6

Anthropic

The daily driver

Fast, capable, excellent coding. Best balance of speed and quality. Powers most Cursor and Claude Code workflows. Your go-to for 80% of tasks.

GPT-5 / 5.2

OpenAI

The enterprise model

Strong reasoning, multimodal, deeply integrated with Microsoft. Powers M365 Copilot, Copilot Studio agents, and Excel analysis. Your model when working in the Microsoft ecosystem.

Gemini 2.5 Pro

Google

The context king

Massive 1M token context window. Strong multimodal. Best when you need to analyze very large documents, codebases, or datasets in a single prompt.

These three models cover 90% of what FM&I needs — learn their strengths and you'll always pick the right one

MODELS SPECIALIST

Specialist & Local Models

Claude Opus 4.6

Anthropic

Deep reasoning powerhouse

The most capable model available. Use when Sonnet isn't enough — complex multi-step analysis, nuanced reasoning, agent teams. Higher cost, slower, but significantly more capable for hard problems.

GPT-4.1

OpenAI

Cost-effective workhorse

Default model in Copilot Studio. Fast, reliable, good enough for most production agent workloads. Use when you need speed and cost efficiency over maximum capability.

Ollama (Local Models)

Various open-source

Full privacy, zero cost

Run models locally on your own hardware. No data leaves your machine. No API costs. Works offline. Use for sensitive data analysis, experimentation, and air-gapped environments.

Opus for hard problems, GPT-4.1 for cheap agents, Ollama for sensitive data — use the right specialist

MODELS EVOLUTION

Model Evolution — The Last 18 Months

Mar 2024	Claude 3 Opus/Sonnet/Haiku launch — first competitive Anthropic models for coding
Jun 2024	Claude 3.5 Sonnet — massive coding improvement, becomes default for many developers
Sep 2024	OpenAI o1 preview — first 'reasoning' model, chain-of-thought for complex problems
Nov 2024	GPT-4o + Claude 3.5 Sonnet v2 — both become go-to models for daily coding work
Feb 2025	Claude 3.7 Sonnet — extended thinking, hybrid fast/deep reasoning
Mar 2025	Gemini 2.5 Pro — Google's best model, 1M token context window
May 2025	Claude 4 Opus + Sonnet — major leap in coding, agent capabilities
Aug 2025	GPT-5 family launches — GPT-5 Chat, GPT-5 Reasoning available
Oct 2025	Claude Opus 4.6 + Sonnet 4.6 — current generation, strongest coding models
Dec 2025	GPT-5.2 in M365 Copilot — improved code generation and multilingual

Models are improving faster than any other technology — what was impossible 6 months ago is routine today

05

Agentic AI & Advanced Concepts

Tool calling, MCP, agent skills, context management — the building blocks of AI agents

ADVANCED AGENTIC AI

The Shift to Agentic AI

Chat AI (2023)

Human asks a question, AI answers. One turn at a time. No memory, no actions, no tools. Think: ChatGPT in a browser window.

Copilot AI (2024)

AI embedded in your workflow — autocomplete, suggestions, summaries. Works alongside you but you drive every action.

Agentic AI (2025)

AI plans, acts, and iterates autonomously. Uses tools, calls APIs, executes code, makes decisions. Human sets goals, agent delivers.

Multi-Agent (2026+)

Teams of specialized agents orchestrated together. A sales agent delegates to a data agent which triggers an analytics agent. Autonomous workflows.

FM&I is entering the agentic phase — the question is what processes can agents run for us autonomously

ADVANCED TOOL CALLING

Tool Calling — How Agents Take Action

What is tool calling? When an AI agent needs to do something beyond generating text — search the web, read a file, call an API, run a database query — it 'calls a tool'. The LLM decides which tool to use, generates the right parameters, the system executes it, and the result flows back into the conversation. This is what turns a chatbot into an agent.

Cursor	MCP servers, built-in tools (file edit, terminal, web search). Agent mode chains multiple tool calls autonomously.
GitHub Copilot	Agent mode uses tools for file creation, terminal commands, workspace search. Extensions add custom tool capabilities.
Claude Code	Native tool use: file read/write, bash execution, web search, MCP servers. Most extensive built-in tool access.
Copilot Studio	Power Platform connectors, MCP servers, Power Automate flows, computer use. 1,400+ pre-built integrations.

Tool calling is what separates 'AI assistant' from 'AI agent' — it's the ability to act, not just think

ADVANCED TOOL CALLING

Tool Calling — Each Tool's Approach

Cursor

MCP-first extensibility

- File read/write, terminal, web search built in
- Full MCP server support — add any URL
- Custom tools via .cursorrules + MCP servers
- Agent mode auto-chains multiple tool calls

MCP ecosystem — connect anything

GitHub Copilot

GitHub-integrated automation

- File edit, terminal, workspace search, GitHub API
- MCP via GitHub MCP Registry (growing fast)
- Extensions + AGENTS.md for custom behavior
- Coding Agent runs async in GitHub Actions

Cloud-based agent runs from Issues → PRs

Claude Code

Full system access by default

- File R/W, bash, web search, notebooks, sub-agents
- Full MCP + custom MCP servers
- Skills system (.claude/skills/) + hooks
- Default mode is fully agentic — deepest access

Deepest built-in tool access of any coding AI

Copilot Studio

Enterprise connectors at scale

- 1,400+ Power Platform connectors built in
- MCP server support (Preview) — one-click setup
- Custom connectors for any REST API
- Autonomous agents triggered by business events

Most enterprise integrations by far

Each tool has a different philosophy — pick based on what you need to connect to

ADVANCED MCP

MCP — Model Context Protocol

What Is MCP?

Model Context Protocol is an open standard (created by Anthropic) that lets AI agents connect to external tools and data sources in a standardized way.

Think of it like USB for AI agents — a universal plug that works with any tool, any agent, any platform.

Why it matters:

Without MCP, every integration requires custom code. With MCP, you build a server once and it works in Cursor, Claude Code, Copilot Studio, and any MCP-compatible agent.

MCP in Practice

- Databricks MCP server — agents query Unity Catalog functions directly
- GitHub MCP — agents create issues, PRs, manage repos
- Slack/Teams MCP — agents send messages, read channels
- Google Calendar/Gmail MCP — agents manage schedule and email
- Custom MCP — build your own for internal APIs and databases
- Copilot Studio — connect MCP servers with one click (Preview)

MCP is becoming the universal connector for AI agents — learn it once, use it everywhere

ADVANCED CONTEXT

Context Windows & Token Management

Why this matters: Every AI model has a finite context window — the total amount of text (code, prompts, history) it can 'see' at once. When you exceed it, the model starts forgetting earlier parts of the conversation. Managing context is the difference between an AI that gives great answers and one that loses the plot halfway through.

Start fresh for new tasks	Don't reuse long conversation threads for unrelated work. Start a new chat — the AI will perform better with focused context.
Use project files instead of pasting	Point the AI at files rather than pasting entire codebases into chat. Tools like Cursor and Claude Code index your project automatically.
Be concise in prompts	Long, rambling prompts waste context. State what you need clearly and let the AI ask clarifying questions if needed.
Summarize before continuing	For very long tasks, ask the AI to summarize progress so far, then continue in a new context with that summary.

Context management is a practical skill — not just a technical concept. It directly affects output quality.

ADVANCED AGENT SKILLS

Agent Skills — Reusable AI Capabilities

What are agent skills? Reusable modules that give AI agents specific capabilities. Think of them like plugins — a skill teaches an agent how to do something: search a database, generate a chart, send an email, run a model. Skills are portable across agents and projects. In Claude Code, skills live in .claude/skills/ and are auto-detected. In Copilot Studio, they're actions and connectors.

Example: Data Quality Check Skill

A reusable skill that takes a dataset path, runs standard quality checks (nulls, outliers, type mismatches, date gaps), and returns a structured report. Any agent can invoke it — the model-building agent, the ETL agent, or an ad hoc analysis agent.

Example: Commodity Price Fetch Skill

Connects to your data source (Databricks, API, CSV), pulls latest prices for specified contracts, normalizes to a standard format. Agents use it whenever they need price data without knowing the data source details.

Example: Model Documentation Skill

Given a model's code and metadata, generates a standardized risk/compliance document following your team's template. Ensures consistent documentation across all FM&I models regardless of who builds them.

Skills turn one-off prompts into reusable team infrastructure — build once, use across all agents and projects

ADVANCED OTHER TOOLS

Hooks, ChatGPT Codex & Other Tools

Claude Code Hooks

- Scripts that run automatically when specific events occur in Claude Code
- Event types: PreToolUse, PostToolUse, Stop, TeammateIdle, TaskCompleted
- Example: auto-run linter after every file edit (PostToolUse hook on Write tool)
- Example: prevent task completion until tests pass (TaskCompleted hook, exit code 2)
- Example: auto-format code before commits (Stop hook runs prettier)
- Hooks enable automated quality gates without manual intervention
- Power user feature — turns Claude Code into a fully automated CI/CD-like pipeline

ChatGPT Codex (OpenAI)

- OpenAI's cloud-based autonomous coding agent — competitor to Claude Code
- Runs in a secure GitHub Actions-based sandbox environment
- Assign it a GitHub Issue → it produces a draft Pull Request autonomously
- Strengths: integration with GitHub ecosystem, async execution
- Key difference from Claude Code: cloud-based (not local), GitHub-centric

Windsurf & Other Entrants

- Windsurf (by Codeium): AI IDE similar to Cursor, strong free tier, Cascade agent
- Amazon Q Developer: AWS-integrated coding assistant, strong for cloud infra
- JetBrains AI: AI features built into IntelliJ, PyCharm — if you use JetBrains already
- The market is moving fast — new entrants monthly, but Copilot, Cursor, and Claude Code lead

Hooks automate quality gates | Codex is OpenAI's cloud coding agent | The ecosystem keeps expanding

COMPLIANCE ETHICS

Ethics & Compliance — Key Principles

Data Classification Matters

Know your data classification before putting it into any AI tool. Confidential and restricted data should never go into external or public AI services without explicit approval.

No Proprietary Data in External Tools

BP proprietary models, trading strategies, and position data must not be shared with external AI services. Use approved internal tools or local models for sensitive work.

Verify AI Outputs

AI models hallucinate. Every output — especially code, calculations, and factual claims — must be verified by a human before being used in any commercial decision.

IP and Licensing

Code generated by AI tools may have licensing implications. Understand your tools' terms of service and BP's IP policy. Don't paste proprietary third-party code into AI tools.

Audit Trail

Keep records of AI-assisted decisions. If an AI helped generate analysis that influenced a trade, document the AI's contribution and your verification steps.

Stay Current

Guidance is evolving rapidly. Check BP's internal AI ethics and compliance portal regularly. When in doubt, ask your line manager or compliance team.

AI is a power tool — use it responsibly. Check BP's internal guidance portal for the latest policies.

COMPLIANCE DATA

Data Classification & Approved Tools

Public / Open

Published data, public filings, open-source code, general knowledge questions

Approved tools: Any tool: ChatGPT, Claude.ai, Cursor, Copilot, external APIs

Internal / General

Internal docs, non-sensitive code, general business processes, meeting notes

Approved tools: Approved enterprise tools: M365 Copilot, GitHub Copilot (Enterprise), internal LLM endpoints

Confidential

Proprietary models, trading strategies, position data, client information, unreleased analysis

Approved tools: Internal only: approved internal API endpoints, Ollama/local models, air-gapped environments

Restricted

Price-sensitive information, MNPI, regulatory submissions, personal data (GDPR)

Approved tools: Do NOT use AI tools. Human review only. Consult compliance before any AI involvement.

When in doubt about data classification, ask your line manager or compliance team BEFORE using AI tools

06

The Business Case for Gen AI

ROI data, value metrics, and a reality check on what the numbers actually mean

VALUE ROI

Gen AI ROI — Hard Numbers & Reality Check

ROI Evidence (Forrester/Microsoft)

- 116% ROI over 3 years, 25K-employee enterprise ($19.7M NPV)
- 132–353% projected ROI for SMBs over 3 years
- 70% of users report higher daily productivity
- 29% faster task completion (presentations, emails, docs)
- 55% coding speed gains (GitHub Copilot users)
- BC Investment Corp: 2,300+ hours saved in pilot
- Commercial Bank of Dubai: 39,000 hours saved annually

Beyond the Numbers

Faster onboarding (25%), reduced burnout, lower attrition (20% projected), 4x faster meeting catch-up, democratized capabilities

The Reality Check

50% of tech leaders can't prove ROI (CNBC). 40%+ struggle to measure impact (Deloitte). Vendor-funded studies may overstate gains. Many pilots stall on governance (Gartner). Start with measurable use cases and track baselines.

Value by Tool Category

M365 Copilot — $30/user/mo	GitHub Copilot — $19-39/mo	Copilot Studio — Consumption	Cursor / Claude — $0-200/mo
70% higher productivity	55% faster coding	Unlimited agents	2-3x faster dev

Start with personal productivity tools now — layer enterprise platforms as org readiness matures

VALUE STRATEGY

Two Worlds of AI — Enterprise vs Personal

Enterprise-Scale AI

6–18 months
to roll out

$500K – $5M+
investment

CTO / CISO / Legal
who decides

Azure OpenAI, Copilot Studio, Bedrock
Centralized governance, SSO, audit, compliance

Personal Productivity AI

Minutes
to install

$0 – $200/mo
investment

You
who decides

GitHub Copilot, Cursor, Claude Code
Individual discretion, low risk, immediate value

Two different worlds — different timelines, different stakeholders, different risk profiles

These aren't competing strategies. They're complementary.

1. **Start with personal tools NOW**
   Install Copilot, Cursor, Claude Code today

2. **Build AI fluency across the team**
   Innovation Day, prompt libraries, template projects

3. **Layer enterprise platforms as org matures**
   Copilot Studio agents, governed workflows, compliance

07

Personal Use Cases & Demos

OpenClaw, Claude tools, and real-world examples of AI in daily life

PERSONAL OPENCLAW

OpenClaw — Your Personal AI Employee

What Is It?

- Free, open-source personal AI assistant (openclaw.ai)
- Runs 24/7 locally on Mac Mini, Raspberry Pi, or VPS
- Connects to WhatsApp, Telegram, Slack, Teams
- 400+ community skills, model-agnostic, you own everything
- Proactive 'heartbeat' messages — morning briefings, reminders

Security Warnings

- Tech preview — creator calls it 'a hobby project'
- 400+ contributors with AI-generated code, minimal oversight
- Broad system access: shell, filesystem, email, messaging
- Prompt injection risk — email could trick the agent
- Mitigation: dedicated device, Docker, don't connect sensitive accounts

Real-World Use Cases

Morning Briefing

Gmail/Calendar/Drive summary

20 min/morning

Code on the Go

Fix tests from phone via Claude Code

Ship without laptop

Email Triage

Monitor inbox, draft responses

30-60 min/day

Worth experimenting to understand the paradigm — but treat as sandbox, not production

PERSONAL OTHER TOOLS

Other Personal AI Tools & Demos

Claude Chrome Extension

AI browsing agent that navigates the web on your behalf. Search, read, interact with websites — all via natural language instructions.

Claude Remote (MCP)

Access your Claude Code sessions remotely. Start a task on your desktop, check progress from your phone. Full remote control of AI workflows.

Remotion + Claude Code

Generate programmatic videos using code. Claude Code writes the Remotion components, you get professional video output. Presentations, explainers, data viz.

Weather Alerts Project

Example personal project: AI monitors weather APIs, sends WhatsApp alerts for conditions that matter to you. Template for any monitoring + notification use case.

Claude Template Projects

Deep dive into project setup with claude.md, agent configurations, and reusable patterns. The _p-presentation-creator project used to build this deck is a live example.

Personal tools are the best way to build AI intuition — experiment freely outside of work first

08

Business Potential & Innovation Day

Copilot for Power BI, Plotly Studio, Dataiku Gen AI, MCP tools, and your ideas

BUSINESS USE CASES

Commercial AI Opportunities for FM&I

Copilot for Power BI

Natural language data analysis directly in Power BI. Ask questions about dashboards, generate DAX formulas, create visualizations with prompts. Transforms how traders consume analytics.

Plotly Studio

AI-powered interactive visualization platform. Build publication-quality dashboards and charts with natural language. Ideal for commodity price analysis, term structures, and correlation matrices.

Dataiku Gen AI Capabilities

Agent Connect (central hub for AI agents), LLM Mesh (multi-model orchestration), no-code agent builder, LangChain integration, AI Code Assistant in Code Studios. Native to our existing platform.

MCP Tools for Dataiku

Build MCP servers that expose Dataiku project outputs, model endpoints, and dataset queries to any MCP-compatible agent — enabling Cursor, Claude Code, and Copilot Studio to interact with our models.

Automated Model Monitoring

Agents that continuously monitor model performance, data drift, and prediction accuracy. Alert the team when models degrade. Shift from reactive to proactive model management.

Natural Language Data Queries

Build agents that let traders ask questions about our data in plain English: 'What was the Brent-WTI spread volatility last month vs 5-year average?' No SQL, no dashboards needed.

Innovation Day goal: pick 2-3 of these and build working prototypes as a team

BUSINESS VISUALIZATION

Copilot for Power BI & Plotly Studio

Copilot for Power BI

- Natural language data analysis directly inside Power BI dashboards
- Generate DAX formulas, create visuals, build reports from prompts
- Ask questions: 'Show me Brent vs WTI spread trend for the last 6 months'
- Summarize datasets, explain trends, identify outliers automatically
- Works with existing Power BI datasets — no migration needed
- Now supports Anthropic reasoning models in Excel integration
- FM&I use: traders query our dashboards conversationally instead of navigating complex filters

Plotly Studio

- AI-powered interactive visualization platform by Plotly (Dash Enterprise)
- Build publication-quality charts and dashboards with natural language
- Supports complex financial visualizations: term structures, vol surfaces, heatmaps
- Connect directly to data sources — no manual data prep required
- Export to interactive web apps, static images, or embed in notebooks
- Python-native: integrates with our existing pandas/scipy workflows
- FM&I use: rapid prototyping of commodity analytics dashboards for trader consumption

Power BI Copilot = AI on existing dashboards | Plotly Studio = AI-built new visualizations | Both serve FM&I

BUSINESS DATAIKU

Dataiku Gen AI Capabilities

Dataiku is already our core platform for ELT pipelines, modelling, and automation. It now has a comprehensive Gen AI layer — Agent Connect, LLM Mesh, AI code assistant, and agent tools — that we can leverage without leaving the platform we already use daily.

Agent Connect

Central hub where users ask any question. Routes to the best agent automatically. All activity tracked for quality monitoring and continuous improvement.

LLM Mesh

Multi-model orchestration gateway. Use OpenAI, Anthropic, Mistral, or self-hosted models. Flexible permissioning, security management, cost controls, and guardrails.

No-Code Agent Builder

Business analysts build agents visually. Developers get full Python and LangChain support. Both share the same enterprise governance layer for security and audit.

AI Code Assistant

LLM-powered code completion inside Dataiku Code Studios via LLM Mesh. Similar to GitHub Copilot but within our Dataiku environment using approved models.

MCP for Dataiku

Build MCP servers exposing Dataiku project outputs, model endpoints, and dataset queries to any MCP-compatible agent — Cursor, Claude Code, or Copilot Studio.

GenAI Registry & Governance

Track every agent and LLM in use. Deployment sign-offs ensure nothing reaches production without review. Value and risk monitoring for all agentic initiatives.

Dataiku's Gen AI layer means we don't have to leave our platform — we extend it with agents and LLMs

BUSINESS

INNOVATION DAY

Innovation Day — Ideas Framework

The goal: Populate an ideas board with use cases people are already using, plus new commercial ideas to explore. Each idea should map to a real FM&I workflow and have a clear path to a working prototype. We're not looking for slideware — we're looking for things we can build and demo in a day.

Quick Wins

Can build in hours. Immediate personal productivity. Start using tomorrow.

Prompt libraries, .cursorrules templates, Copilot workflows for common tasks

Team Enablers

Build in days. Shared tools that help the whole squad. Reusable infrastructure.

MCP servers for our data, agent skills library, shared prompt collections

Commercial Impact

Build in weeks. Directly affects trading decisions or commercial delivery speed.

Automated model monitoring, NL data queries for traders, risk doc generation

Add your ideas to the Mural board — we'll vote and assign teams on Innovation Day

AI is making the cost of work approach zero.

The question isn't 'can we build this?' anymore. It's 'should we?' and 'who owns the outcome?'

When code is cheap, judgment becomes the scarce resource. Domain expertise — understanding commodity markets, trading dynamics, and commercial risk — is what makes AI output valuable instead of just fast.

AI won't replace data scientists. But tool users will replace non-tool users.

The next decade won't kill technology jobs — it will kill the comfortable middle.

People who confuse shipping software with owning outcomes will be the most exposed.

AI replaces the IT middle layer that never understood the subject domain — medicine, law, commodities.

The performance art of coding is dying: requirements ceremonies, slideware architecture, digital transformation slogans.

The big question: who owns the risk? Not who wrote the prompt.

The people who will thrive are the ones who understand the domain AND the tools.

That's FM&I. We understand commodity markets. Now let's master the tools.

NEXT STEPS

TRY IT

What You Can Build This Week

30 min

Document Your Project (Claude Code)

```
'Read all files in [your project dir] and generate a README.md documenting the project structure, key functions, data flow, and dependencies.'
```

You'll get a comprehensive README that would have taken an hour+ to write manually. If it works, try: 'Now generate model documentation for compliance.'

1 hour

Refactor Your Worst Function (Cursor)

```
Paste your most complex function into Composer and type: 'Refactor this for readability, add type hints, comprehensive docstrings, and write 5 unit tests.'
```

You'll see a multi-file diff: cleaner code, type safety, tests, docs — all generated. Review and accept. If it works, try a multi-file refactor.

2 hours

Build a Dashboard from Data (Claude Code)

```
'Build a Streamlit app that loads [your CSV], shows descriptive stats, plots the time series with a date range selector, and highlights anomalies > 2 std dev.'
```

A working interactive dashboard from one prompt. Deploy locally, share the URL with your team. If it works, try adding correlation analysis between commodities.

Pick ONE challenge. Do it TODAY. Time yourself. Then tell the team what happened.

Gen AI for Front Office Analytics

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Questions | Ideas | Let's Build

Resources: BP AI Ethics Portal | Internal Copilot Training | Tool Request Portal | Innovation Day Mural Board